(a) Tactile description

Paired fingertip signals · last T frames

P2L Left tactile clip

P2R Right tactile clip

t − T + 1 … t

Shared spatial CNN

Temporal convolution

Field classifiers

Discrete tactile text

area · Fx · Fy · Fz · rotation

Reference: left-finger sensor frame

(b) Shared reasoning and action model

Tactile semantics

C Common inputs

P1F Front · P1W Wrist

Current 7D qpos

Task instruction

T Tactile text

Current Touch

H Motion history

qpos H100 → 11

P Current plan

none if unavailable

F Global failures

Current + past failures

Shared $\pi_{0.5}$ / PaliGemma backbone

Hold & reason

On need = true

C T H P

Failure diagnosis

Global failure sequence

C F

Recovery planning

Direction + magnitude

Execution → need = true: hold → plan → **Adjustment**

adj = true

Repeated assessment

Fresh observation

Select by phase

need Execution C T H P

adj Adjustment C T H P

false: resample

true?

true → interrupt

update phase

One chunk · repeated checks

false … true

Action generation & motion

Action Expert

execution C

adjustment C P

H30

Executing

P3 Robot interaction

(c) Asynchronous execution–recovery loop

Monitor during motion · interrupt within a chunk

need = true · end = true

Execute · Hold & reason · Recover · Resume task

Motion: HOLD

Assessment: need_recovery · Diagnose → plan · adjustment_end · need_recovery

t

Resume, re-evaluate, and recover again if a fault remains.

Solid: information / action flow Dashed: feedback / stage control • Shared-model calls are serialized; monitoring overlaps physical motion.